

Spring 2025

Math F251X

Calculus 1: Midterm 2

Name: _____

Section: ☐ 9:15am (James Gossell)

☐ 11:45am (Kevin Meek)

☐ async (Deven Barnett)

Rules:

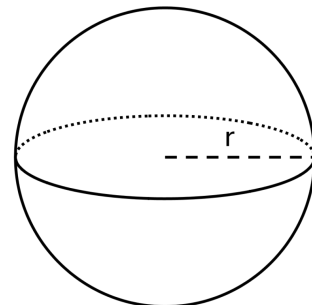
- Partial credit will be awarded, but you must show your work.
- You may have a single handwritten $3'' \times 5''$ notecard, both sides.
- Calculators are **not** allowed.
- Place a box around your FINAL ANSWER to each question where appropriate.
- Turn off anything that might go beep during the exam.

Good luck!

Problem	Possible	Score
1	9	
2	6	
3	16	
4	12	
5	12	
6	12	
7	10	
8	9	
9	8	
10	6	
Extra Credit	(5)	
Total	100	

1. (9 points)

The volume of a sphere is given by $V = \frac{4}{3}\pi r^3$ where r is the radius of the sphere. If the radius is decreasing at a constant rate of 2 meters per second, find the rate at which the sphere's volume is decreasing when the radius is 5 meters. Include units in your final answer.

**2. (6 points)**

Let $g(x) = x + \cos(x)$. Determine the absolute maximum and absolute minimum of $g(x)$ on the interval $[0, 2\pi]$. Justify your conclusion with work.

maximum value of $g(x)$: _____

minimum value of $g(x)$: _____

3. (16 points)

Let $g(x) = x^{4/3} - 8x^{1/3}$. Note that we have the following:

$$g'(x) = \frac{4x-8}{3x^{2/3}} \qquad g''(x) = \frac{4x+16}{9x^{5/3}}$$

You must show your work for all parts to receive credit!

- Determine the intervals where g is **increasing** and where g is **decreasing**.
- Find the x values where any **local maxima** occur and where any **local minima** occur.
- Find the intervals where g is **concave up** and where g is **concave down**.
- Find the x values where any **inflection points** occur.

4. (12 points)

Evaluate the following limits. **Show your work**, including appropriate use of limit notation. If you use L'Hôpital's rule, you must indicate where you are using it by writing $\stackrel{H}{=}$ or $\stackrel{L'H}{=}$ or something similar. Use ∞ or $-\infty$ where appropriate, and if the limit does not exist, write DNE and provide a justification.

a. $\lim_{t \rightarrow \infty} \frac{2t^2 - 3t}{12 - 5t^2}$

b. $\lim_{\theta \rightarrow 0} \frac{4 \cos(\theta) - 4}{e^\theta - \theta - 1}$

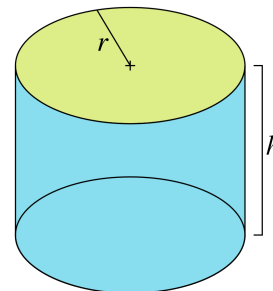
c. $\lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\ln(x^2)}$

d. $\lim_{x \rightarrow 0} x \cot x$

5. (12 points)

The surface area of a closed cylinder is given by $S = 2\pi r^2 + 2\pi rh$ and the volume of a closed cylinder is given by $V = \pi r^2 h$ where r is the radius and h is the height of the cylinder. A packaging scientist is asked to design a cylindrical can with volume $36\pi \text{ cm}^3$ with the **smallest possible surface area**.

- a. Write the expression to be minimized in terms of one variable.



- b. Using calculus, find the radius and height of the cylinder with the smallest surface area. Justify that your answer gives the absolute minimum surface area.

The dimensions of the cylinder with the minimum surface area are:

radius = _____ height = _____

6. (12 points)

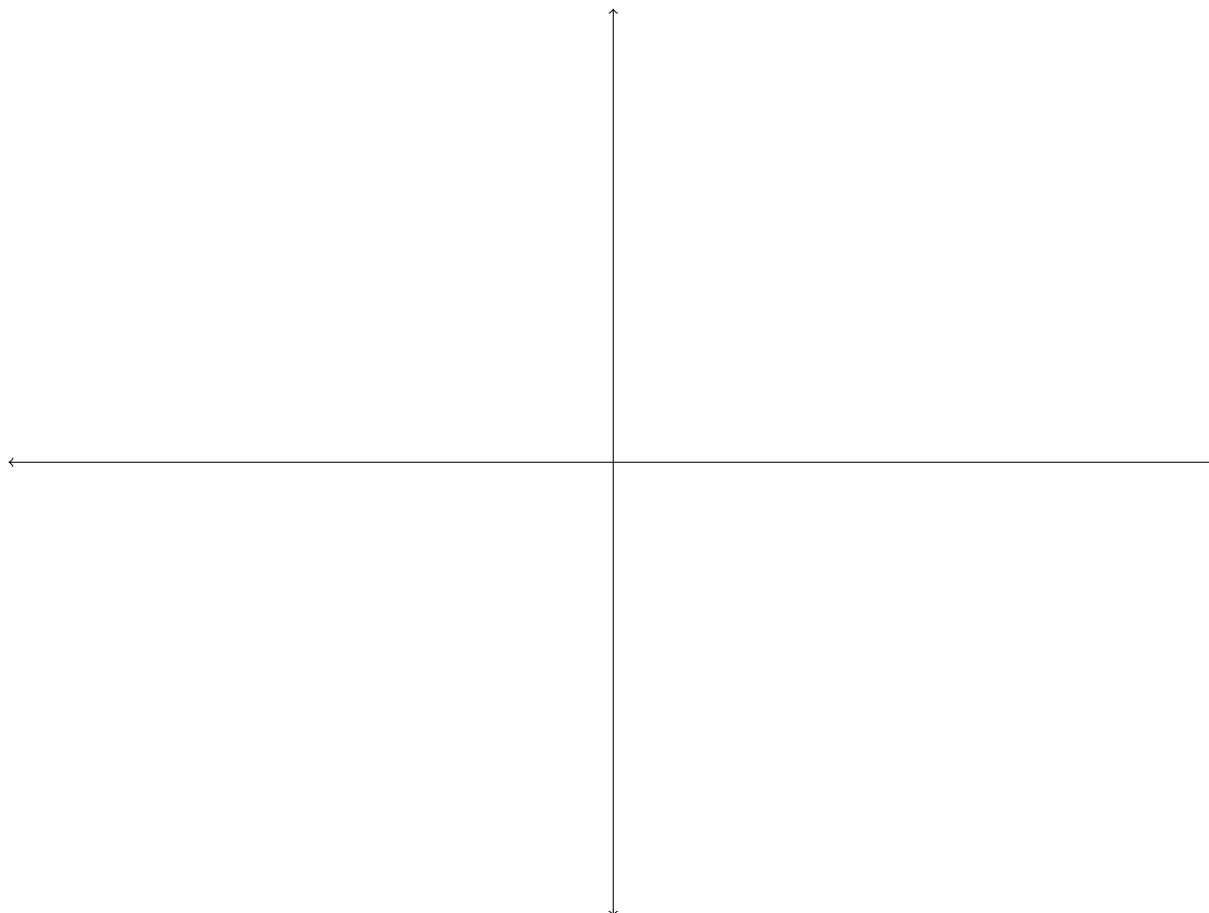
Sketch a graph of a function $f(x)$ that satisfies all of the following properties.

After drawing the graph:

- **Label** on the graph the following things, if they exist, by drawing a point on the graph and labeling: any local maximums by writing LOCAL MAX, local minimums by writing LOCAL MIN, inflection points by writing IP
- Draw any horizontal and vertical asymptotes with dashed lines and **label** them with their equation.
- Mark any important x -values and y -values on the x - and y -axes.

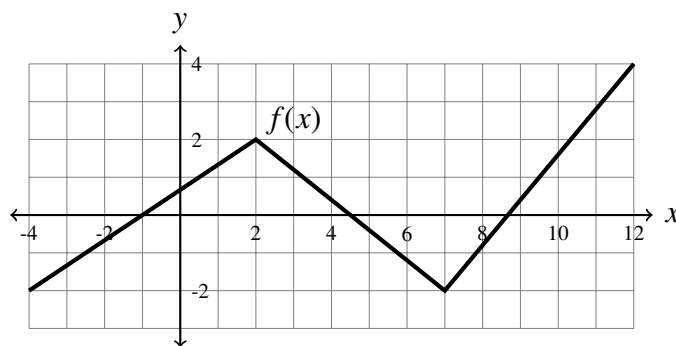
Properties:

- Domain is $(-\infty, 4) \cup (4, \infty)$
- $f(1) = -1$, $f(-3) = 2$
- $\lim_{x \rightarrow \infty} f(x) = 0$
- $\lim_{x \rightarrow -\infty} f(x) = 3$
- $f'(x) > 0$ on $(0, 4)$
- $f'(x) < 0$ on $(-\infty, 0) \cup (4, \infty)$
- $f''(x) > 0$ on $(-2, 4) \cup (4, \infty)$
- $f''(x) < 0$ on $(-\infty, -2)$



7. (10 points)

Consider the graph of the function $f(x)$ shown below:



a. Determine $\int_{-1}^7 f(x) dx$. Show some work or say something about what you computed.

b. Determine $\int_{-1}^7 (2f(x) + 3) dx$. (Hint: Use part (a) above.)

8. (9 points)

Compute the following **indefinite integrals** (antiderivatives). Give the most general answer, and show your work using correct notation.

a. $\int \sqrt[5]{x^3} + 3e^x - \cos(x) \, dx$

b. $\int x^2(3x + 4) \, dx$

c. $\int \frac{1 - 2x + 4x^2}{x} \, dx$

9. (8 points)

A drone is released into the air from an initial height of 5 feet off the ground. Its upward velocity at t seconds is given by $v(t) = \frac{1}{1+t^2}$ feet per second. *(Recall: $\frac{d}{dx} \arctan(x) = \frac{1}{1+x^2}$).*

a. How high will the drone be after 1 second? Please include units in your answer.

b. Compute $\lim_{t \rightarrow \infty} v(t)$, and write a sentence which interprets your result in the context of the problem.

10. (6 points)

Suppose we want to estimate $\sqrt{4.1}$.

- a. Find the linear approximation (linearization) of the function $f(x) = \sqrt{x}$ when $a = 4$.

- b. Use the linear approximation from part (a) to estimate $\sqrt{4.1}$. Your answer must be in the form of a simplified decimal or an exact fraction.

11. (Extra Credit: 5 points)

Compute $\lim_{x \rightarrow \infty} \sqrt{x^2 - 1} - x$. **You must show your work.**